



Investigating Patterns of Covariation in Great Egret Neck Motion During Predatory Strikes

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INTRODUCTION

Hérons and egrets (Ardeidae) are wading-birds that capture prey with a rapid strike driven by coordinated head and neck movements. Most species forage by slowly walking or standing motionless while visually locating fishes and other aquatic prey before executing a rapid projection of the head and bill toward the target.

The necks of herons and egrets are highly flexible, multi-jointed systems capable of rapid and precise forward propulsion of the head during prey-capture strikes. Understanding how motion is coordinated across the neck is essential for explaining how these rapid feeding movements are produced. Because prey capture directly influences energy acquisition and ultimately fitness, understanding how predators achieve rapid and accurate strikes is an important ecological question.



Figure 1. Example video frames of a Great Egret (*Ardea alba*) prey-capture strike

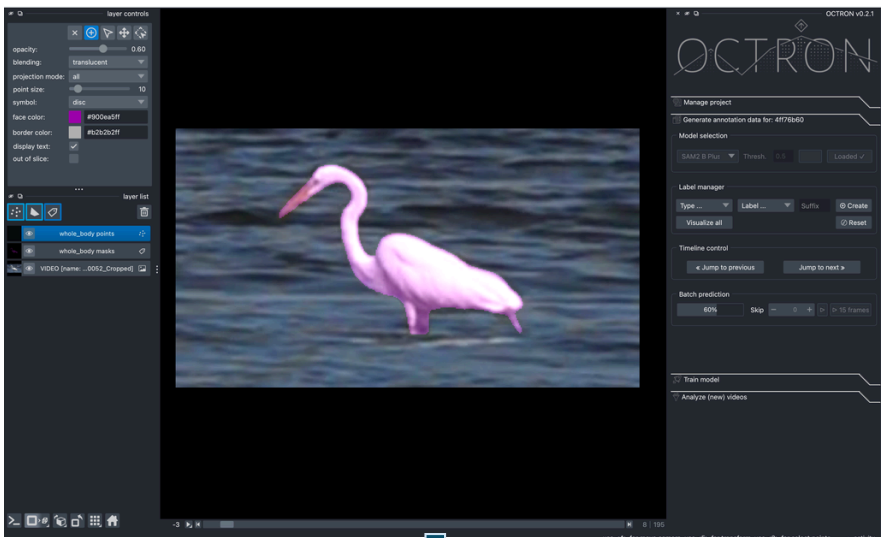
Here, I present preliminary data testing whether different regions of the Great Egret neck function as coordinated modules during strikes. Specifically, I ask whether adjacent segments move together in distinct groups, or whether motion is distributed more continuously along the cervical spine. Characterizing how a multi-jointed biological system coordinates motion across numerous degrees of freedom is a fundamental biomechanical question with implications for understanding motor control in vertebrates.

Main Question: Do different regions of the Great Egret neck function as distinct functional modules during prey-capture strikes?

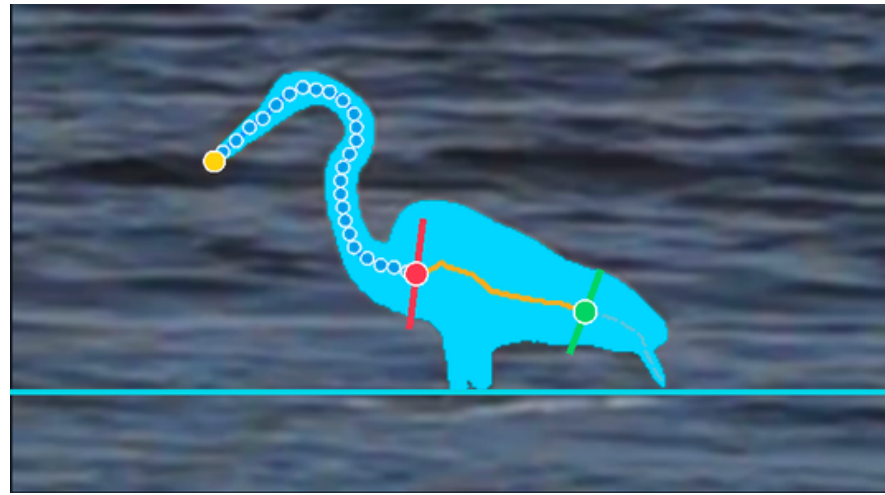
METHODS



Original Frame



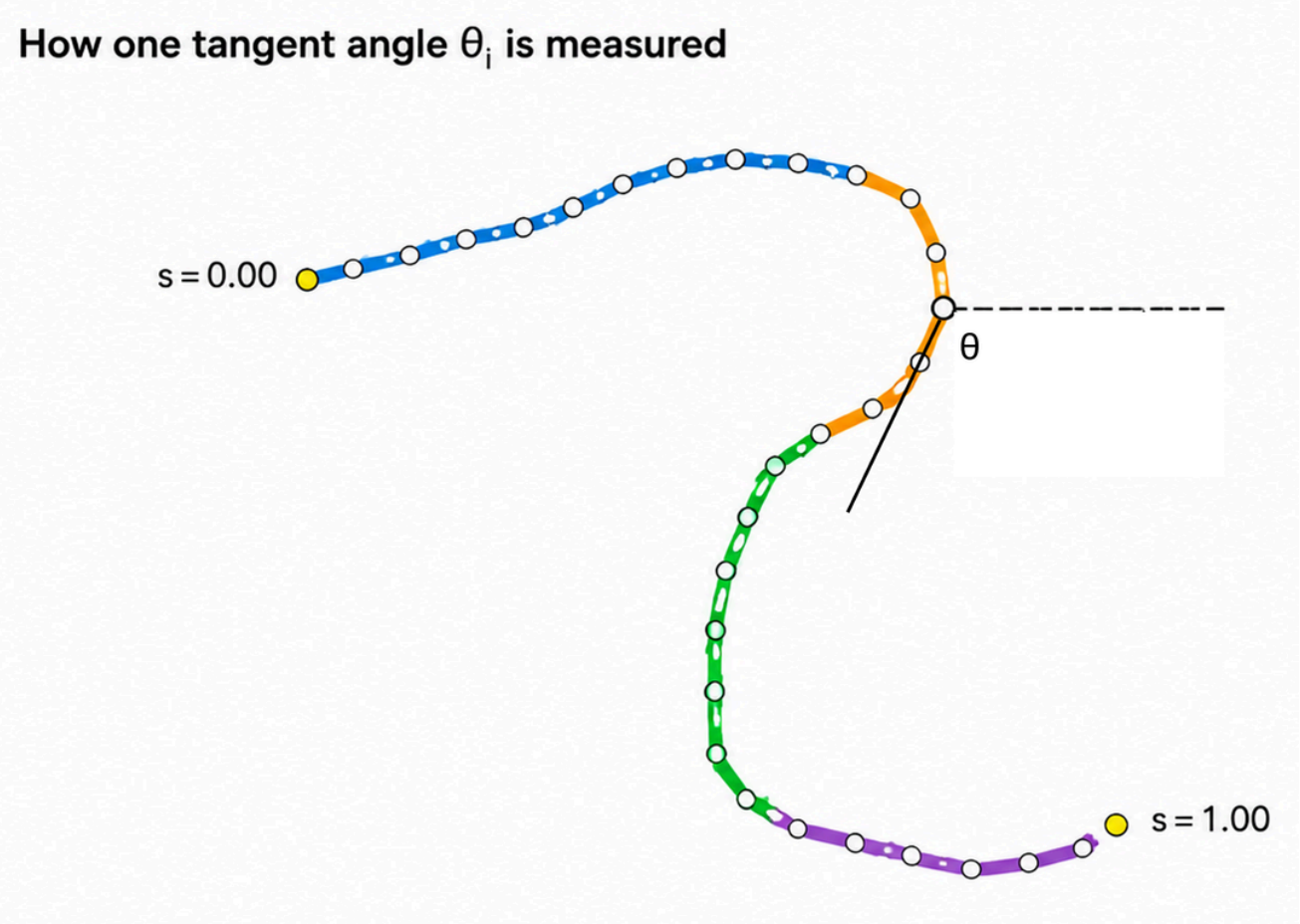
Whole-Body Mask



Midline Extraction



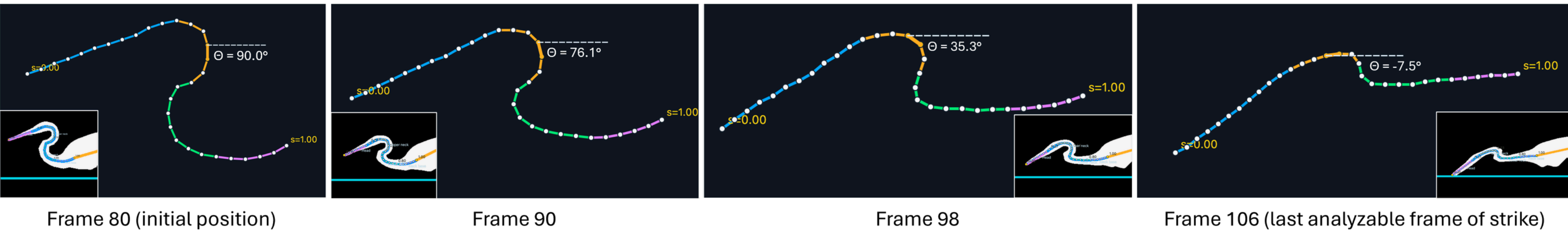
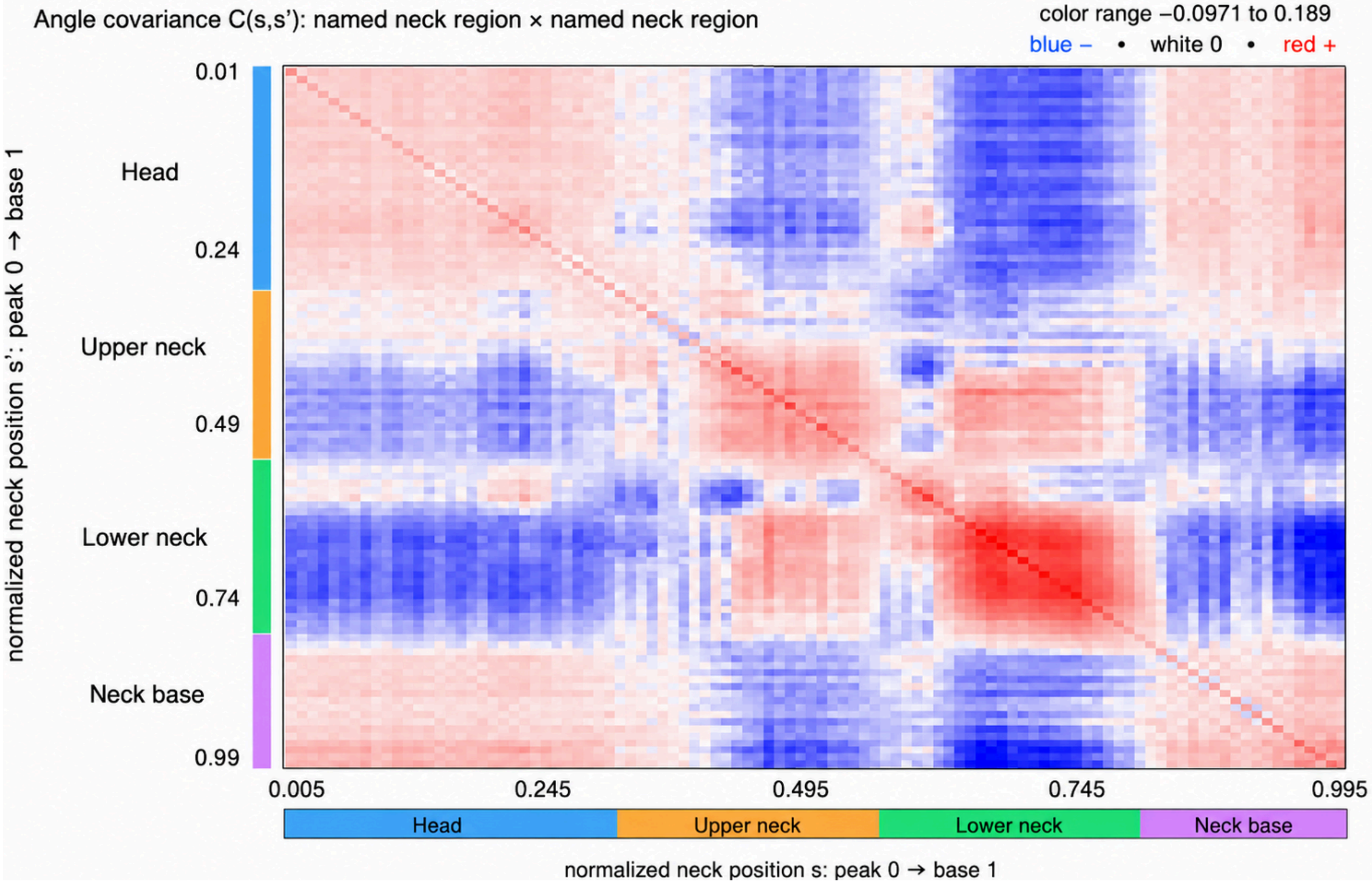
Segmentation & Tangent angles



For each segment, a tangent vector is computed from two adjacent centerline points. The tangent angle (θ) is defined as the angle between this vector and the horizontal axis.

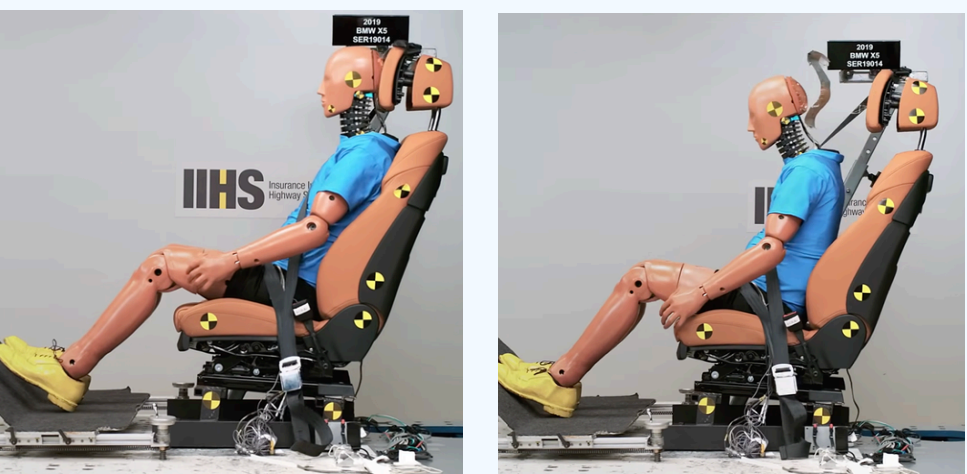
Mean tangent angle across all segments is subtracted from each frame to remove overall neck orientation.

PRELIMINARY RESULTS

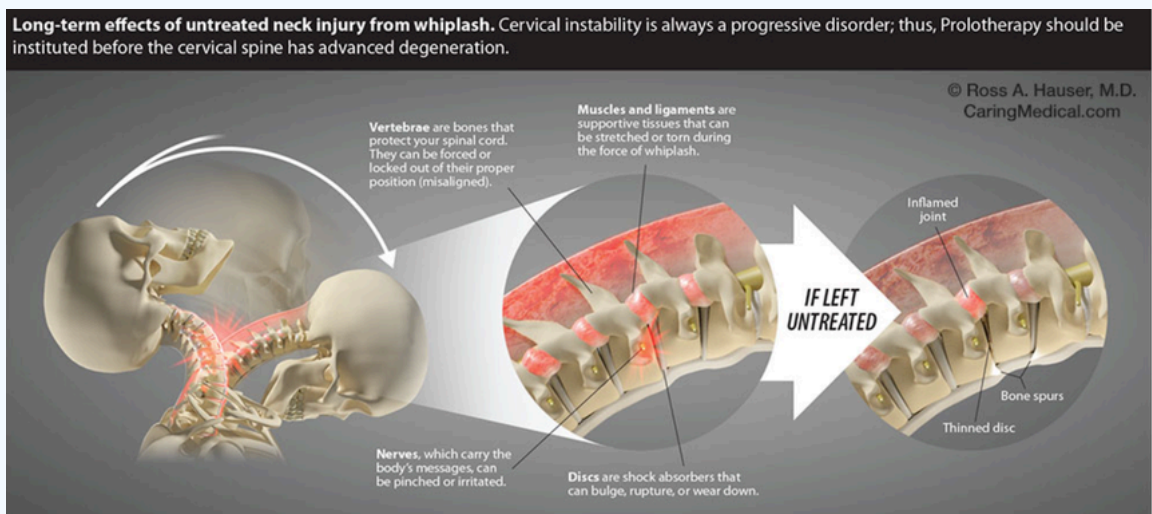


POTENTIAL IMPLICATIONS

Future studies may determine whether principles of neck coordination observed in biological systems can inform bio-inspired engineering or rehabilitation



Active head restraint systems (e.g., BMW) deploy during rear-end collisions to reduce head lag. The modular coordination observed in the Egret neck raises questions about how multi-segment support might influence load distribution during rapid deceleration: an open area for future study.



Schematic of progressive cervical spine changes associated with untreated whiplash injury. The image illustrates how ligament damage, disc thinning, and vertebral misalignment can lead to chronic instability and joint degeneration over time (Hauser, 2023). This contrasts with the Great Egrets, which perform repeated high-acceleration neck strikes without apparent long term injury, which raises comparative questions about how modular coordination might influence load distribution and tissue integrity.

KEY TAKEAWAYS & FUTURE DIRECTIONS

1. Covariance patterns suggest the presence of distinct functional modules during prey-capture strikes.
2. Both positive and negative covariation occur among neck regions, indicating that different regions contribute differently to strike motion.
3. These findings are based on a representative strike and will be evaluated across additional strikes.
4. Next step: Estimate head position after submergence to quantify neck motion throughout the complete strike.

REFERENCES AND VIDEOS

